

Predictive Targeting and Segmentation of Food-Insecure Households in the U.S. FoodAPS Survey

EE 5290 - Final Project Report

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Abstract

Food-assistance organizations face increasing demand under constrained funding and capacity. Using a cleaned, household-level extract of the USDA FoodAPS survey (4,826 households, 38 variables), we develop a two-track data-analytics framework that helps such organizations **(i)** predict which households are food insecure and **(ii)** discover *types* of food-insecure households whose profiles suggest distinct outreach strategies. We compare L2-regularized logistic regression, an RBF-kernel SVM, and a random-forest classifier under three class-imbalance strategies, and we use PCA followed by K-means clustering, both with and without geographic features, to expose household personas. On the held-out 20% test set, logistic regression achieves PR-AUC 0.568 and ROC-AUC 0.783, with a top-decile precision of 67% capturing 24% of food-insecure households. Geography-free persona clustering identifies five interpretable groups; 88% of the model's top-decile flags fall into two of them (*non-working low-income singles* and *large low-income families with children*), suggesting that targeted outreach can focus effort on programs designed for these distinct populations.

1. Introduction

Food security - reliable access to nutritious food - is a foundational component of household well-being. Across the Midwest in particular, demand for assistance is rising while organisational capacity is not, forcing food banks and outreach groups to make explicit choices about *which* households to prioritize. This project asks two related questions on the cleaned FoodAPS-derived dataset provided for the case:

whether the household is food insecure with enough accuracy to be useful for triage?

of households exist, and how should that affect outreach?

- **Prediction.** Given household characteristics, can we predict
- **Segmentation.** Among food-insecure households, what *kinds*

We address both questions with library-first scikit-learn modelling, focusing depth on (i) a controlled comparison of three classifiers under three class-imbalance strategies, (ii) a careful choice of K for K-means based on silhouette and stability, and (iii) a synthesis section that joins the model's risk score to each household's persona to produce concrete targeting recommendations.

Contributions.

recodes survey sentinel codes, encodes mixed nominal/ordinal features, and adds log-transformed access variables.

and random forest on imbalanced food-insecurity prediction with operating-point analyses suited to a triage workflow.

geography-driven view and a five-persona, geography-free view.

model's top-decile to top-half operating points.

- A reusable preprocessing pipeline for the FoodAPS extract that
- A head-to-head evaluation of logistic regression, RBF-kernel SVM
- Two complementary clusterings of the food-insecure subset: a
- A synthesis showing how the persona mix changes across the

Non-causal framing. The dataset is observational and a survey sample. All findings are reported as *associations* between features and the food-security outcome, never as causal effects.

2. Background and Methods

We summarize the four learning methods used in the report; full derivations are standard in the course textbooks.

Logistic regression. Given features x in $\mathbb{R}^d$ and a binary outcome y in $\{0,1\}$, the model $\Pr(y=1|x) = \sigma(w^T x + b)$ with $\sigma(z) = 1/(1+\exp(-z))$ is fit by minimizing the negative log-likelihood with L2 regularization controlled by C . The coefficients w are interpretable on standardized inputs: $w_j > 0$ raises the predicted probability when feature j increases.

Kernel SVM. The dual form of the support vector machine replaces inner products $x_i^T x_j$ with a kernel $k(x_i, x_j)$. We use the radial basis function (RBF) kernel $k(x_i, x_j) = \exp(-\gamma ||x_i - x_j||^2)$ with γ controlling locality and C the soft-margin penalty. Probabilities are produced by Platt scaling on cross-validated margins.

Random forest. Ensembles of decision trees fit on bootstrap samples with feature-bagging give a strong nonlinear baseline that captures interactions automatically and yields impurity-based feature importances.

PCA and K-means. PCA computes the top k singular vectors of the centered standardized data matrix; the projection retains the directions of maximum variance. K-means partitions samples into k clusters by alternating assignment to the nearest centroid and recomputation of centroids; we choose k by inspecting the elbow of the within-cluster sum-of-squares (inertia) jointly with the silhouette score.

Class imbalance. With a positive-class rate near 28%, accuracy is misleading: a trivial all-negative classifier achieves 72% accuracy. We therefore optimize and report *average precision* (PR-AUC) and supplement it with *recall at fixed precision* thresholds. We compare three strategies: no adjustment, balanced class weights (cost-sensitive loss), and probability-threshold tuning chosen on out-of-fold training predictions.

3. Data and Preprocessing

The provided dataset (data/Case_dataset.csv) contains 4,826 households described by 38 columns covering food-security outcomes, economic conditions, household composition, employment and demographics, food-assistance program participation, transportation and access indicators, and food-acquisition behavior. The binary outcome is `food_insecure_flag_adult`; 27.85% of the rows are positive (see figures/01_class_balance.png). The four-level `adltfscat` and the household-reported `foodsufficient` are excluded from the feature matrix to avoid label leakage.

3.1 Cleaning decisions

Survey sentinel codes (-996, -997, -998) appear in `foodpantry`, `anyvehicle`, `vehiclenum`, and `caraccess`; we recode them to NaN before any analysis. Two columns are dropped entirely: `caraccess` (92% sentinel-coded) and `fah_storetype_unique` (mean ~ 0.002 , near-constant). String-coded categorical heads (`head_hispanic`, `head_racecat`) carry the response code "R" for "refused"; we recode "R" to NaN. The survey weight `hhwgt` is documented but not used in modelling (see Section 7).

3.2 Feature engineering

Distance variables (`dist_sm`, `dist_walmart`, `nearsnap_dist` and similar) are right-skewed; we *add* a $\log(1+x)$ transform alongside the raw value, so that linear models see a near-symmetric signal while tree models retain the original units. Four household-composition derivations are added: children per adult, elderly share, child share, and employed-adult share. Nominal categoricals (`head_sex`, `head_hispanic`, `head_racecat`, `head_employment`, `region`, `poverty_band`, `targetgroup`) are one-hot encoded with NaN mapped to an explicit "unknown" level. Numeric NaN are median-imputed. The resulting matrix has 63 features after encoding.

3.3 Train/test split and EDA highlights

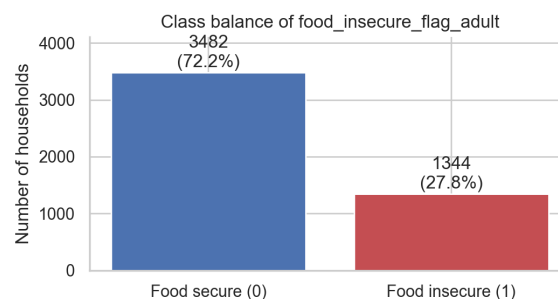
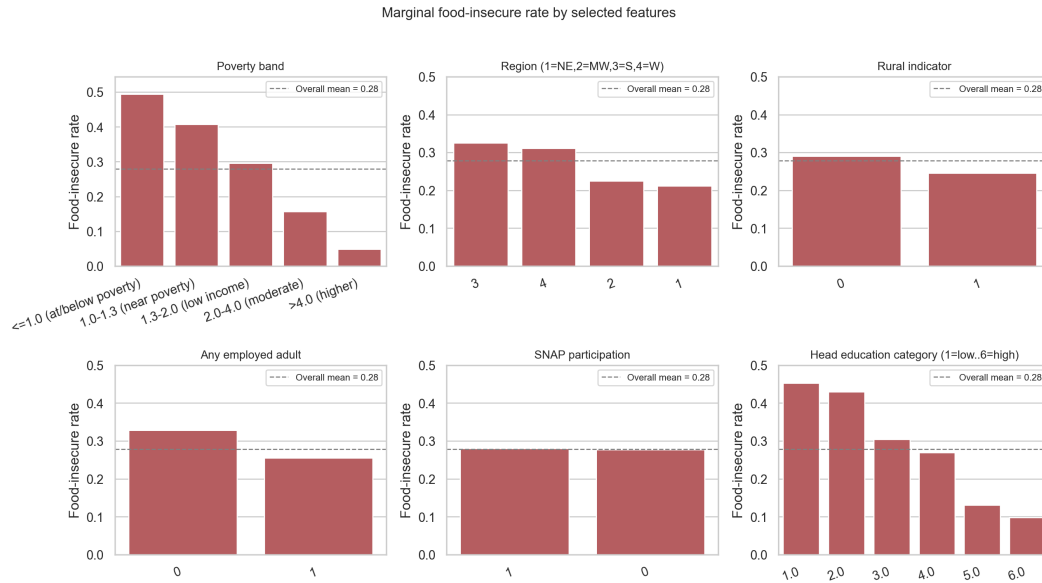


Figure: Class balance and marginal rates.



We hold out a stratified 20% of households as a final test set (966 rows) using `random_state=42`. All hyperparameter tuning uses 5-fold stratified CV on the training fold only; the test set is touched exactly once at the end of Section 4. See `figures/01_class_balance.png` and `figures/02_marginal_rates.png` for class balance and marginal food-insecurity rates by selected features. Households at or below the poverty guideline have nearly a 50% positive rate; the rate is also visibly elevated for SNAP-using households and households without an employed adult. Bars are descriptive only and do not control for confounding.

4. Track 1: Predictive Modelling

We compare three classifiers under three imbalance strategies for a total of nine evaluated configurations. For each base classifier we run `GridSearchCV` over a small grid (LR: C in {0.01, 0.1, 1, 10}; SVM-RBF: C in {0.5, 1, 5}, gamma in {scale, 0.01, 0.1}; RF: max_depth in {None, 10, 20}, min_samples_leaf in {1, 5} with 300 trees) using 5-fold stratified CV on the training fold and the average-precision (PR-AUC) score function.

4.1 Threshold-tuning protocol

For each model, we use `scikit-learn`'s `cross_val_predict` to generate *out-of-fold* probability estimates on the training rows. We then choose the probability threshold that maximizes F1 on those out-of-fold predictions. This avoids using the test set to choose the threshold, which would inflate the reported metrics.

4.2 Test-set results

Model (strategy)	thr	ROC-AUC	PR-AUC	Brier	Prec.	Rec.	F1	R@P=.50
LR - no adjust	0.50	0.7831	0.5681	0.161	0.624	0.364	0.460	0.610
LR - balanced	0.50	0.7829	0.5681	0.195	0.463	0.762	0.576	0.598
LR - F1-tuned thr	0.289	0.7831	0.5681	0.161	0.466	0.755	**0.576**	0.610
SVM-RBF - no adjust	0.50	0.7551	0.5290	0.171	0.549	0.208	0.302	0.565
SVM-RBF - balanced	0.50	0.7677	0.5269	0.166	0.564	0.346	0.429	0.651
SVM-RBF - F1-tuned	0.236	0.7551	0.5290	0.171	0.425	0.747	**0.542**	0.565
RF - no adjust	0.50	0.7667	0.5208	0.166	0.602	0.186	0.284	0.524
RF - balanced	0.50	0.7665	0.5101	0.179	0.485	0.665	0.561	0.658
RF - F1-tuned thr	0.301	0.7667	0.5208	0.166	0.453	0.792	**0.577**	0.524

Three observations stand out:

0.7831 and PR-AUC 0.5681. RF and SVM-RBF reach PR-AUC ~ 0.52 . This is consistent with a problem whose signal is largely linear in the engineered features (a coincidence we exploit in 4.4).

substantially better than the default 0.5 threshold** for the F1/recall combination. Both strategies trade roughly 17% of precision for an extra 30-50 percentage points of recall. The operationally relevant message is that the choice of threshold matters more than the choice of class-weight scheme.

precision floor, the best model recovers only $\sim 26\%$ of the food-insecure population. This sets a realistic expectation for the targeting recommendation in Section 6.

- **Logistic regression wins on every aggregate metric: ROC-AUC**
- **Class weights and threshold tuning are nearly equivalent and
- **Recall at high precision is small for every method: at a 70%**

4.3 Visual diagnostics

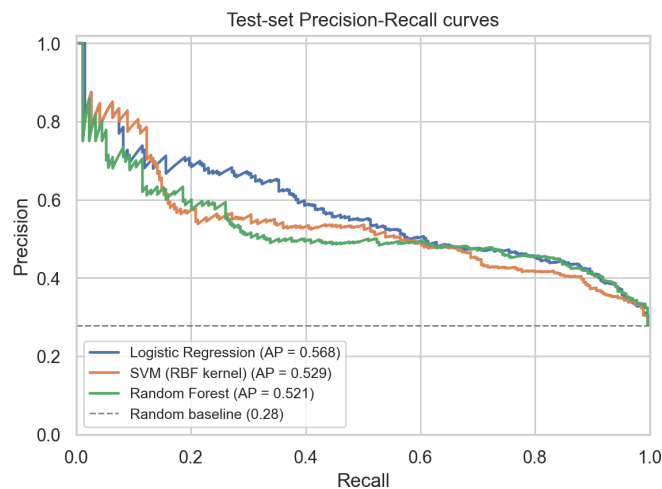
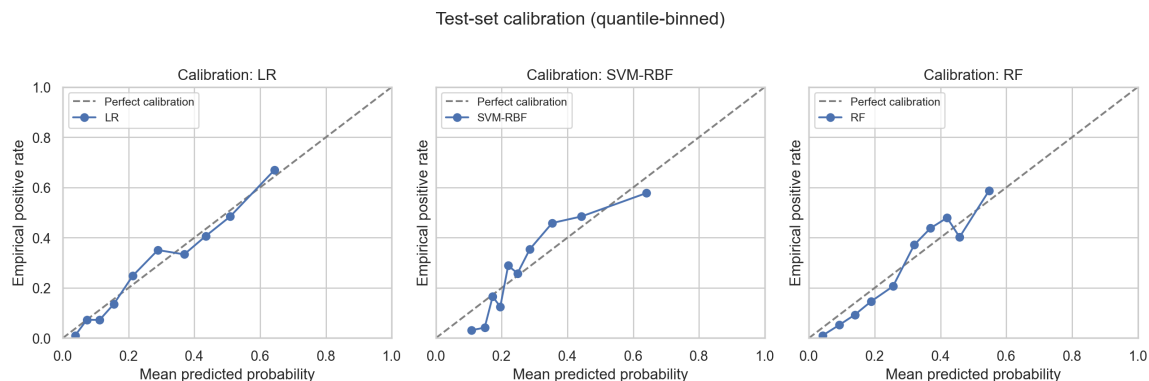


Figure: PR curves and calibration.



figures/05_pr_curves.png shows the precision/recall curves for the three classifiers' raw probability outputs. LR's curve dominates SVM and RF in the high-recall regime, the regime that matters for a food bank that wants to maximize coverage of the food-insecure population subject to a precision floor. figures/06_calibration.png shows the calibration diagnostics: LR is well calibrated across the range; the random forest is mildly over-confident at the extremes; the SVM is well-behaved in the middle but slightly under-confident at the low end (an artefact of Platt scaling on a limited sample).

4.4 Interpretation

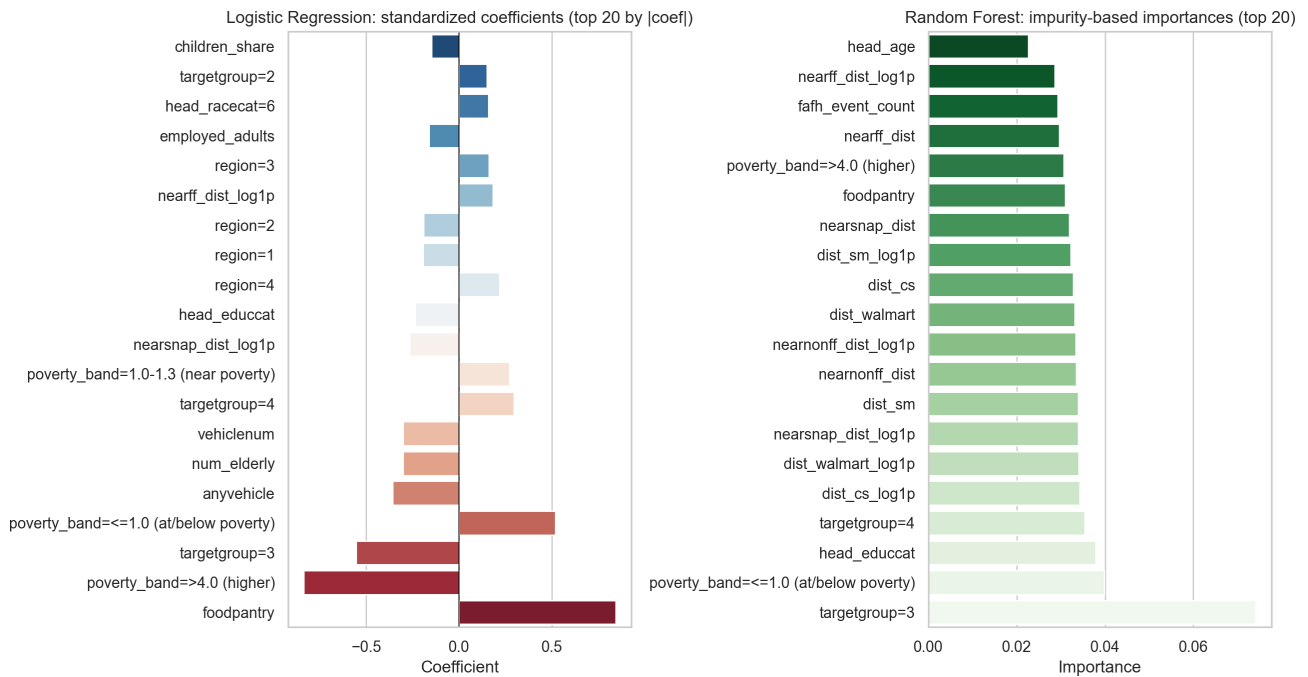


Figure: Top-20 feature importances (LR coefficients vs RF importances).

See figures/08_feature_importance.png. The two interpretation views are remarkably consistent. Food-pantry use, low poverty band, and the FoodAPS sampling target group dominate both rankings; the random forest additionally elevates the cluster of distance variables, reflecting that geographical access is the largest source of nonlinear variation even after controlling for income. Logistic regression - which can report *direction* - shows expected protective effects of more vehicles, higher education, and elderly household members (the latter likely reflects greater enrollment in income-stabilising programs such as Social Security and SNAP). We use foodpantry as a predictor because the data dictionary lists it under "program participation"; we acknowledge it is a behavioural correlate of insecurity rather than an independent causal driver.

5. Track 2: Segmentation of Food-Insecure Households

The 1,344 food-insecure households are projected onto principal components of the standardized 63-feature matrix; 24 components explain 80% of the variance. We then sweep K-means for k in $\{2..8\}$ with 20 random restarts per k .

5.1 Geographic clustering: a strong rural/urban axis

The silhouette criterion identifies $k=2$ (silhouette = 0.221) as the clear winner; $k=3..8$ all hover near 0.10. The two clusters correspond cleanly to rural ($n=222$, mean dist_walmart ~ 11 miles) and non-rural ($n=1,122$, mean dist_walmart ~ 3 miles) households. Stability across eight random initializations is high (mean ARI > 0.9). This is a real but uninformative finding for outreach planning: it tells the food bank that physical access dominates within-population variation but not what programmes to design within each access tier.

5.2 Geography-free persona clustering

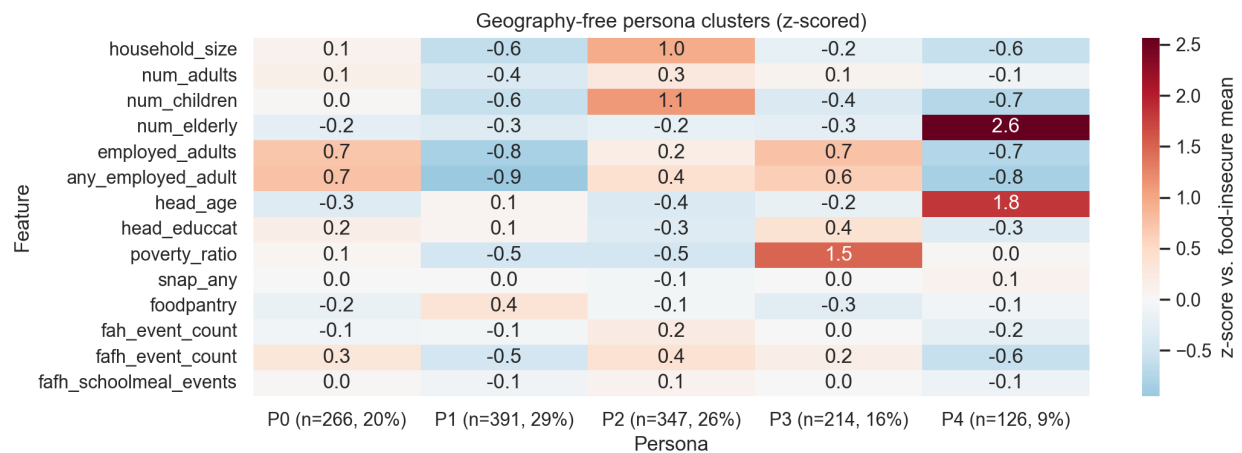


Figure: Geography-free persona z-scores (k=5).

We therefore re-cluster after dropping the seven geographic features (distances, rural, nonmetro, region one-hots). The remaining 50 features capture household composition, employment, demographics, poverty, program use, and food-acquisition behaviour. Silhouette is maximized at k=2 as before but more usable values appear at k in {3,4,5}; we cap at k=5 for narrative tractability and report that solution. See figures/14_persona_heatmap.png.

The five personas admit clean narrative names:

food-insecure subset). Nearly all have an employed adult; poverty ratio 1.43; moderate household size; pantry use only 9%.

29.1%). Small households (mean size 2.0); only 16% have an employed adult; poverty ratio 0.85; pantry use 28% - triple the food-insecure average. This is the cluster most clearly served by direct food distribution.

Mean household size 5.0 with 2.6 children; poverty ratio 0.85; partial employment; lower head education. School-meal eligibility is highest in this cluster.

15.9%). Highest poverty ratio (3.05, well above the poverty guideline) and highest education; near-universal employment. These are households whose food insecurity is unlikely to be resolved by income-eligibility programmes.

9.4%). Mean head age 70; lowest employment; highest SNAP enrolment (42%). Their inclusion in the food-insecure subset suggests SNAP is necessary but not always sufficient.

- **P0 - Working families just above poverty** (n=266, 19.8% of the
- **P1 - Non-working low-income singles, high pantry use** (n=391,
- **P2 - Large low-income families with children** (n=347, 25.8%).
- **P3 - Working educated households near the boundary** (n=214,
- **P4 - Elderly low-income households, high SNAP enrollment** (n=126,

6. Synthesis: Targeting Recommendations

We assign every household in the test set to a persona by passing it through the same preprocessing -> behaviour-only PCA -> K-means pipeline used in Section 5.2, and rank households by the LR model's predicted probability. This lets us look at *who* the model would actually be flagging at each operating point.

See figures/15_persona_mix_at_cutoffs.png and figures/16_precision_recall_vs_cutoff.png.

Top-decile composition. At the top 10% predicted-risk cutoff, precision is 68% (97 of the 966 test households are flagged; 66 are truly food-insecure, capturing 24% of the test set's food-insecure population). Of those 97 flagged households, **45% are P1 (non-working low-income singles) and 43% are P2 (large low-income families with children)**. The two clusters whose risk the model rates lowest - P3 (working educated near boundary) and P4 (elderly on SNAP) - together account for under 2% of the top-decile flags despite making up over half the population.

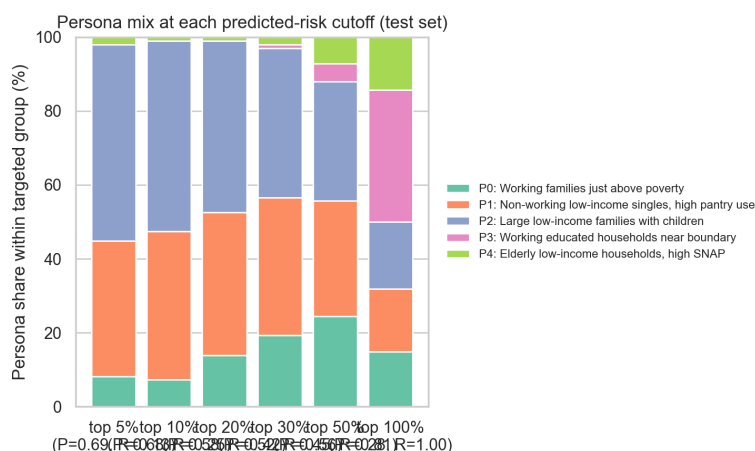


Figure: Persona composition at each top-X% cutoff.

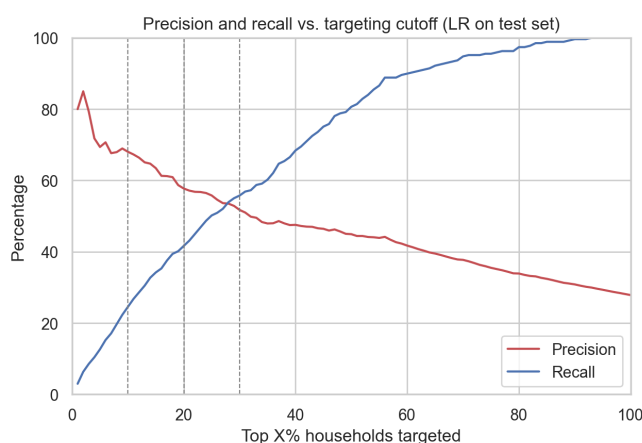


Figure: Precision and recall vs. cutoff.

Operational implication. A food bank that adopts top-decile triage of the model can plan outreach as if the workload would split evenly between two distinct programme tracks: (a) supplemental food distribution for low-income adult singles (potentially via existing pantry partnerships, since this group already over-uses pantries); and (b) family-oriented programmes for large low-income households with children, including school-meal coordination and bulk-pack distributions. The remaining 12% of the top decile sits in P0 (working families just above poverty) and warrants light-touch SNAP-enrolment outreach.

What the model misses. Persona P3 (working educated households just above the poverty line) is 36.9% of the population but only 15.9% of the food-insecure subset; the model rates them with mean probability 0.12 and they account for <0.5% of the top-decile flags. These households may experience episodic insecurity that the survey's snapshot does not capture well, and they are unlikely to be reached by income-eligibility programmes. This is a population that warrants qualitative follow-up rather than algorithmic targeting.

Equity inspection. Persona P4 (elderly on SNAP) is under-represented in the top decile relative to its share of food-insecure households (1.0% versus 9.4%). Two interpretations are consistent with the data: the model correctly de-prioritizes households whose insecurity is partly already addressed by SNAP and Social Security; or the model under-weights this group because the snapshot fails to capture elderly-specific access barriers (mobility, dietary restrictions). The recommendation is to monitor this group via existing SNAP partnerships rather than treat the model's low scores as a green light to disengage.

7. Limitations and Future Work

Survey weights ignored. The provided dataset is a probability sample with weights in hhwtg. We did not use them in fitting or in evaluation, on the basis of the data dictionary's note that weights are optional. A weighted retrain - using sample_weight in the LR loss and a weighted scoring function - would produce different coefficient magnitudes and

may shift cluster sizes; we recommend this as the first follow-up.

Observational data. Every coefficient and importance is an *association*, not a causal effect. In particular, the strong positive coefficient on foodpantry is best read as "households that already use pantries are also reporting food insecurity", not "pantry use causes food insecurity". The same caveat applies to snap_any: its protective sign here cannot be read as a causal estimate of SNAP's effectiveness.

Imbalance methods left unexplored. We compared no adjustment, balanced class weights, and threshold tuning, and found them roughly equivalent in F1 with no method dominating across precision floors. Future work should add SMOTE or its variants for synthetic minority oversampling and cost-sensitive losses with explicit cost matrices, and should evaluate them on a Bayesian decision-theoretic objective that weights false-negative cost (missing a food-insecure household) against false-positive cost (wasted outreach effort).

Persona stability across outcome thresholds. Our personas are fit on households flagged on the binary outcome. Refitting on the 4-level adltfscat (low vs. very low food security) and on near-poverty households who narrowly avoided the binary flag would test whether the personas track *insecurity* or *poverty*.

External data. The case allows supplementing with USDA county-level access data and ACS demographics. Adding even a single county-level summary (e.g., median income, supermarket density) would plausibly tighten the model and let us sanity-check distance features against published food-desert maps.

Reproducibility. All results are reproduced by notebooks/01..04.ipynb with random_state=42 and requirements.txt pinned to scikit-learn 1.8. See README.md for setup.

8. Conclusion

A library-first analysis of the FoodAPS extract is sufficient to build both a useful predictive model (PR-AUC 0.57, top-decile precision 67%) and an interpretable five-persona segmentation of the food-insecure population. The synthesis of the two - reading off *who* the model flags at the top decile - turns an abstract ranking into concrete programme recommendations: roughly half the top decile is non-working low-income singles, half is large low-income families with children, and the remaining 12% is working families on the poverty boundary. This kind of analysis is exactly the type of decision support that capacity-constrained food banks need; the methodology generalizes to other household-survey datasets with mixed nominal/ordinal features and an imbalanced binary outcome.

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